ECM3412/ECMM409
Nature Inspired Computation
Lecture 5

Evolutionary Computation

Encodings / Applications

Contents

- Representation examples
- Problems with Crossover
- Direct & Indirect Encodings
- Innovation in EAs

Encoding / Representation

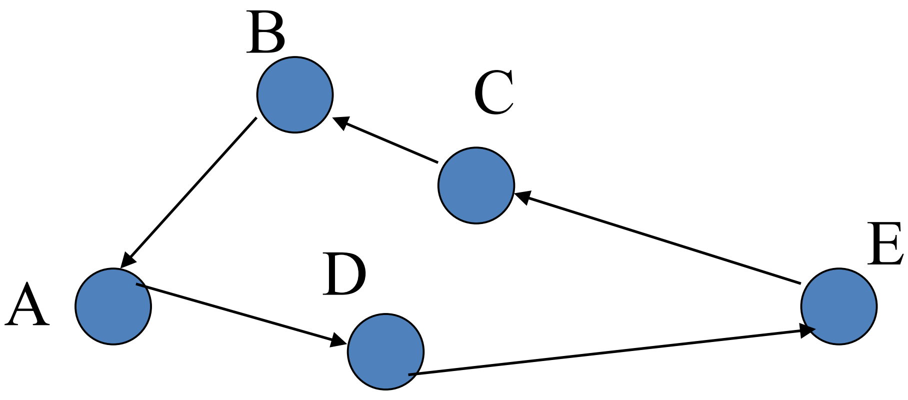
Maybe the main issue in (applying) EC

Note that:

- Given an optimisation problem to solve, we need to find a way of encoding candidate solutions
- There can be many very different encodings for the same problem
- Each way affects the shape of the landscape and the choice of best strategy for climbing that landscape.

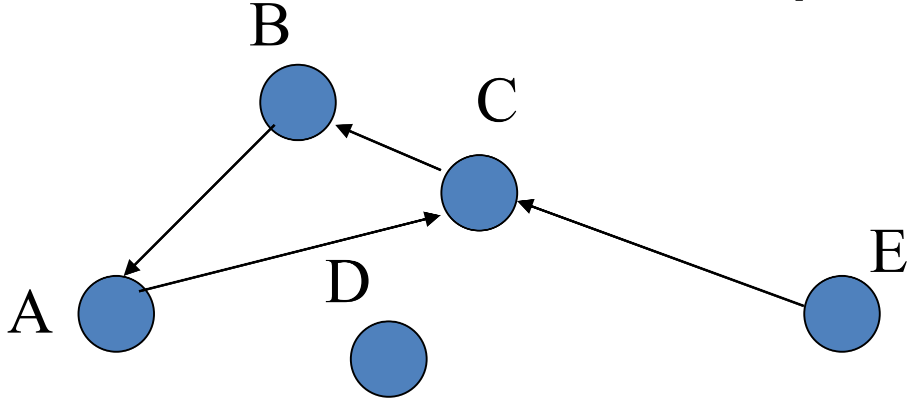
Representation Example

- Revisiting the TSP



A	D	E	C	B

- Standard Mutation (randomly change one of the values in the chromosome):

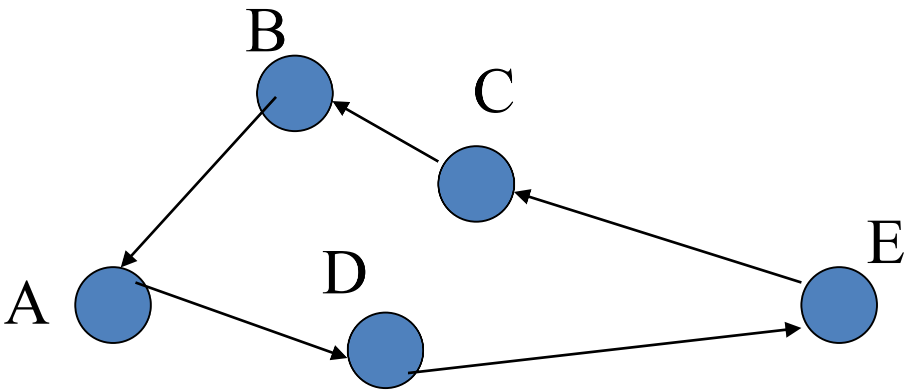


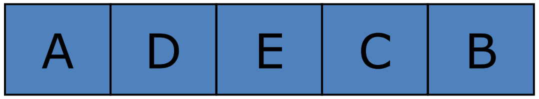
A	C	E	C	B

So we now visit C twice and D not at all!!

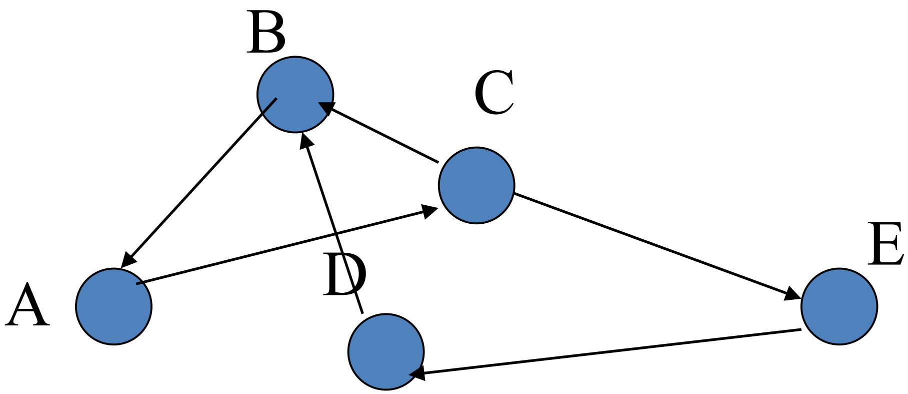
Representation Example (contd.)

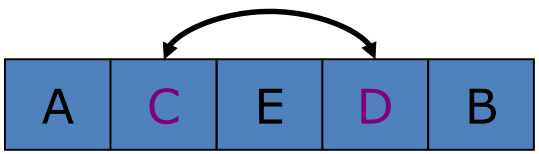
- This is clearly not what is intended, so we must replace the standard mutation with something which does the job properly.





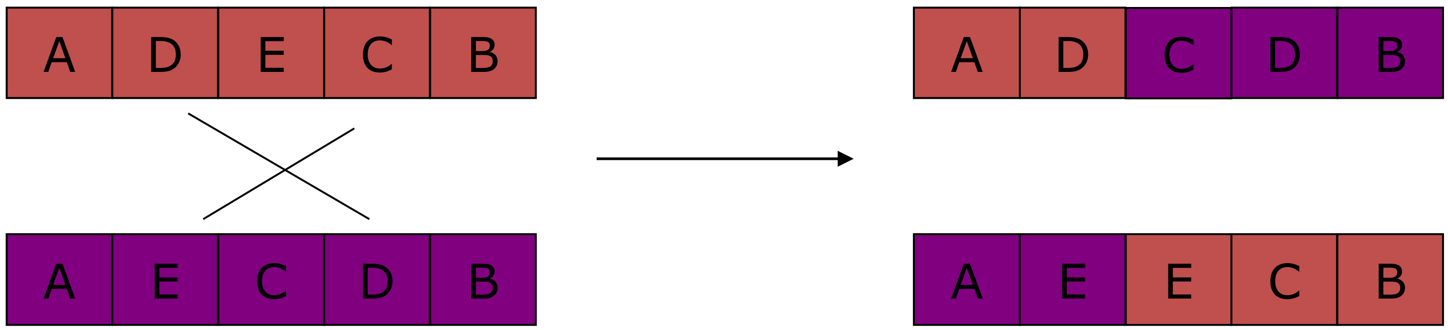
- The swap operator would seem to work:





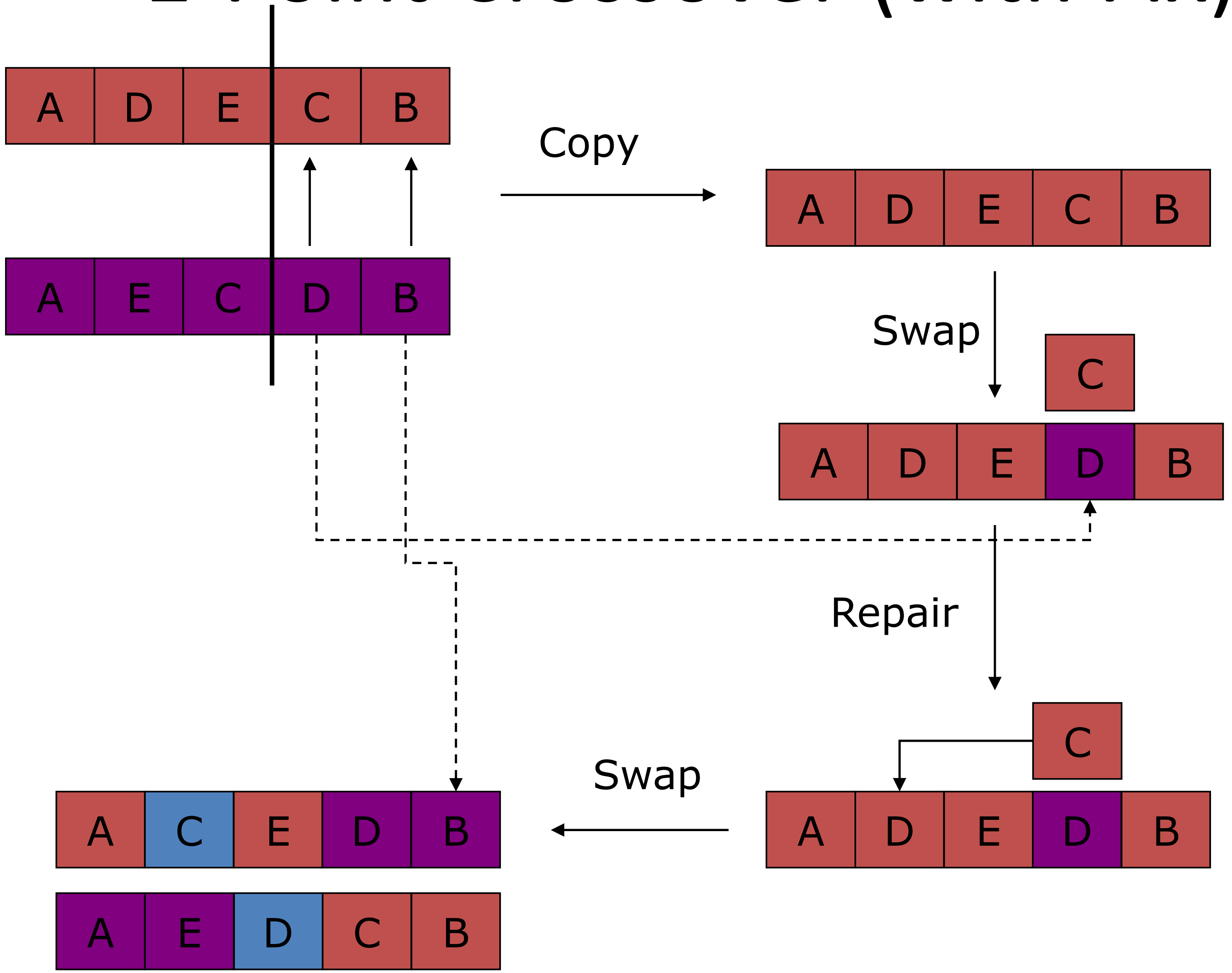
Representation Example (contd.)

- The same problem exists with crossover.
- Standard crossover



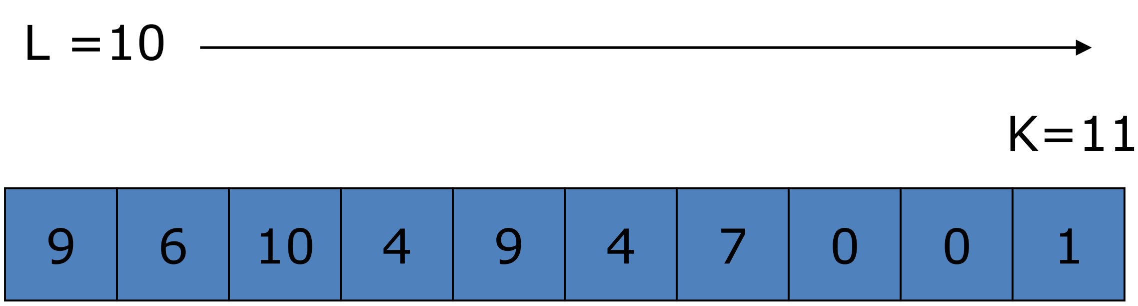
- This gives 2 invalid children which visit the same node twice
- 2 Strategies can help:
  1. Have a better cross-over in the first place
  2. Introduce a ‘fix’ to make the crossed-over solutions valid again.

1-Point Crossover (with Fix)



Some Simple Direct Encodings

- Water Distribution Network Optimisation
  - There are *n* pipes with *m* possible diameters.
  - The problem is to create a new network design from these variables.
  - Each pipe must have a diameter, so the encoding can be a k-ary encoding: a chromosome of *l* integers long with 1->*k* possible values



  - Actual diameters (e.g. inches, mm etc… are sourced from a lookup table)

Some Simple Direct Encodings

- Generalised Assignment Problem
  - If you have *n* workers and *m* jobs to complete, what encoding would you use?

  Constraints:
  - Every job must have exactly one worker assigned to it.
  - Each worker can do more than one job, but each job takes a specified amount of the workers (finite) time (different workers takes different times for the same job).

  Objective:
  - We want to minimise the total overall time to complete all jobs.

Encodings

Encoding 1

- encoding of *n* integers (workers), each with a range 1-*m (jobs)*
  - \+ every worker has at least one job
  - \- not every job will have a worker assigned
  - \- cannot assign more than one job to a worker
  - \- one job may be assigned to 2 or more workers

n $\longrightarrow$

m

J9	J6	J3	J4	J6	J7	J1	J8	J8	J6

0

J2? J5?

Encoding 2

- encoding of *m* integers (jobs), each with a range 1-*n (workers)*
  - \+ every job has a worker assigned to it
  - \- a worker may be overworked

m $\longrightarrow$

n

W9	W6	W3	W4	W6	W7	W1	W8	W8	W6

0

E.g. encoding a timetable I

4, 5, 13, 1, 1, 7, 13, 2

Exam2 in 5th slot

Exam1 in 4th slot

Etc …

	mon	tue	wed	thur
9:00	E4, E5	E2		E3, E7
11:00	E8			
2:00		E6		
4:00	E1			

- Generate *any* string of *8* numbers between 1 and 16, and we have a timetable!
- Fitness may be <clashes> + <consecs> + etc …
- Figure out an encoding, and a fitness function, and you can try to evolve solutions.

Mutating a Timetable with Encoding 1

4, 5, 13, 1, 1, 7, 13, 2

	mon	tue	wed	thur
9:00	E4, E5	E2		E3, E7
11:00	E8			
2:00		E6		
4:00	E1			

- Using straightforward single-gene mutation
- Choose a random gene

Mutating a Timetable with Encoding 1

4, 5, 6 , 1, 1, 7, 13, 2

	mon	tue	wed	thur
9:00	E4, E5	E2		E7
11:00	E8	E3		
2:00		E6		
4:00	E1			

- Using straightforward single-gene mutation
- One mutation changes position of one exam

Mutation with an Indirect Encoding

4, 5, 10, 1, 1, 7, 15, 2

Etc …

	mon	tue	wed	thur
9:00	E4 E5	E2		
11:00			E3	
2:00	E8	E6		E7
4:00	E1			

Use the 10th clash-free slot for exam3

Use the 5th clash-free slot for exam2

Use the 4th clash-free slot for exam1

Suppose these groups would clash (e.g. **non-consecutive**, non-parallel)

{E1, E2}, {E1,E3}, {E2, E6}, {E2,E7}, {E2,E8}, {E3,E5}, {E3,E6},
{E4,E6}, {E4, E7}, {E5, E7}, {E5, E8}, {E6, E8}

Mutation with an Indirect Encoding

4, 5, 10, 1, 1, 7, 15, 2

	mon	tue	wed	thur
9:00	E4 E5	E2		
11:00			E3	
2:00	E8	E6		E7
4:00	E1			

Use the 10th clash-free slot for exam3

Use the 5th clash-free slot for exam2

Use the 4th clash-free slot for exam1

Suppose these groups would clash

{E1, E2}, {E1,E3}, {E2, E6}, {E2,E7}, {E2,E8}, {E3,E5}, {E3,E6}, {E4,E6}, {E4, E7}, {E5, E7}, {E5, E8}, {E6, E8}

Mutation with an Indirect Encoding

4, 5, 10, 1, 14, 7, 15, 2

	mon	tue	wed	thur
9:00	E4	E2		
11:00	E8		E3	E5
2:00		E6		
4:00	E1			E7

Use the 13th clash-free slot for exam3

Use the 5th clash-free slot for exam2

Use the 4th clash-free slot for exam1

Suppose these groups would clash

{E1, E2}, {E1,E3}, {E2, E6}, {E2,E7}, {E2,E8}, {E3,E5}, {E3,E6},
{E4,E6}, {E4, E7}, {E5, E7}, {E5, E8}, {E6, E8}

Direct vs Indirect Encoding

Representation/Encoding		Modifies	Invalid Solutions
Direct	1,2,3,4 011010 ABFCDE	A variable in the problem E.g. Exam Time, Water Pipe Size	Dealt with solely by penalising fitness
Indirect	1,2,3,4 011010 ABFCDE	The variables of a constructive heuristic E.g. Clash-Free Exam Time, Rules for Sizing Water Pipes	Mostly dealt with by encoding. Some penalising by fitness if necessary

Direct vs Indirect Encodings

Direct:

- straightforward genotype (encoding) → phenotype (individual) mapping
- Easy to estimate effects of mutation (smoother landscape)
- Fast interpretation of chromosome (hence speedier fitness evaluation)

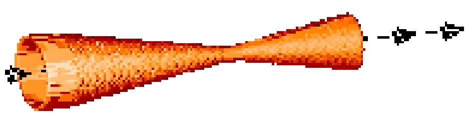
Indirect:

- Easier to exploit domain knowledge
- Hence, possible to enforce problem constraints
- Hence, can seriously cut down the size of the search space
- But, slow interpretation
- Neighbourhoods are highly rugged.

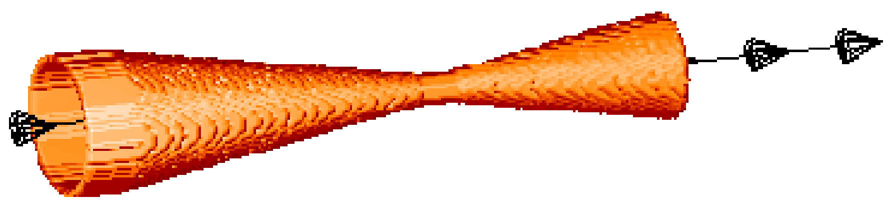
One of the very first applications. Determine the internal shape of a two-phase jet nozzle that can achieve the maximum possible thrust under given starting conditions

Ingo Rechenberg was the *very* first, with pipe-bend design

This is slightly later work in the same lab, by Schwefel

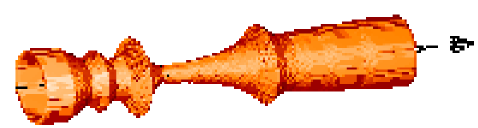


Starting point



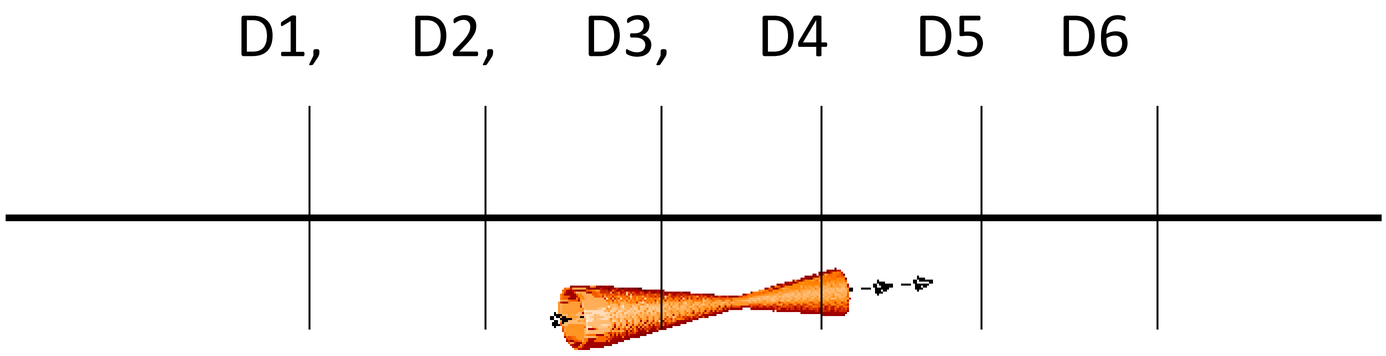
EA (ES) running

Result



A recurring theme: design freedom → entirely new and better designs based on principles we don't yet understand.

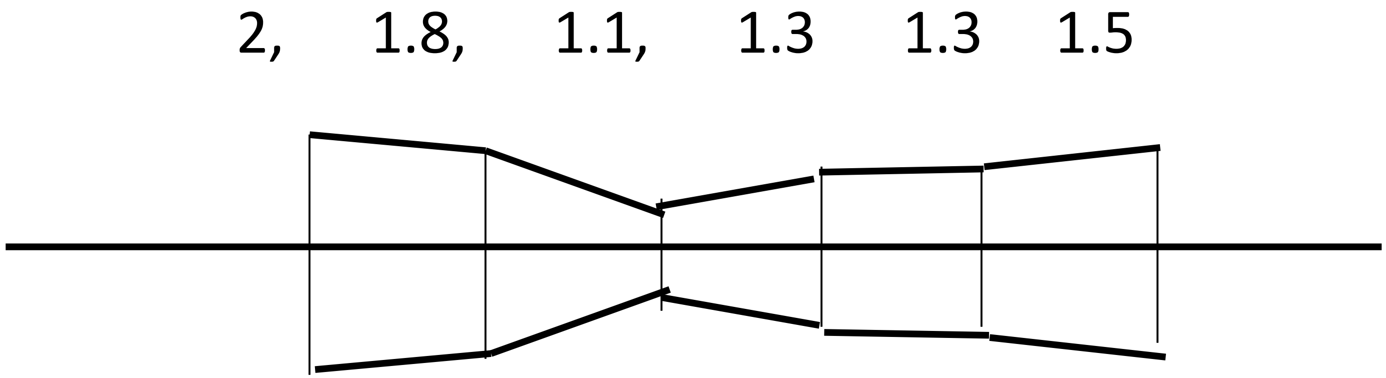
A Real Encoding (and: How EAs can innovate, rather than just optimize)



D1 >= D2 >= D3, D4 <= D5 <= D6

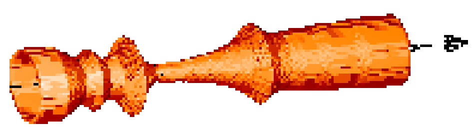
Fixed at six diameters, five sections

An example genotype and its phenotype



Assume each is allowed to vary between 0.1 and 2.

What is the likely effect of *random-gene* mutation where we replace a gene with a random new value in the range?



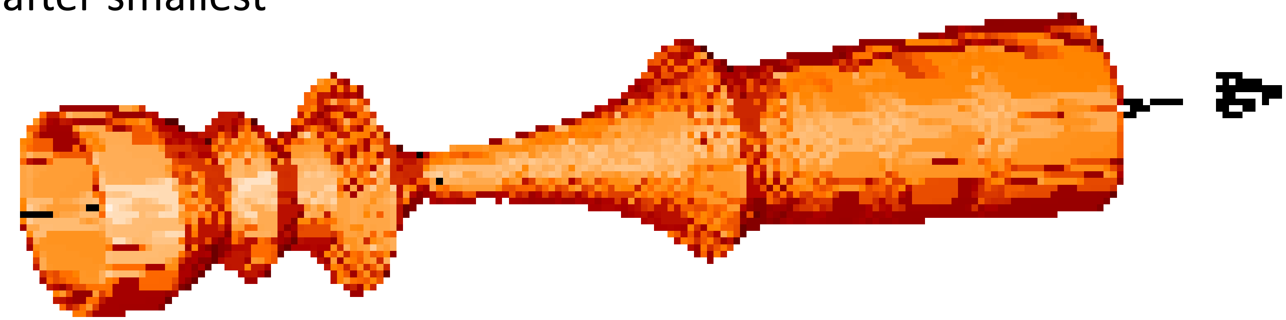
E.g. How EAs can innovate, rather than just optimize

Num sections before smallest

Section diameters

Z1, Z2, D1, D2, D3…, Dsmall, Dn, Dn+1, …

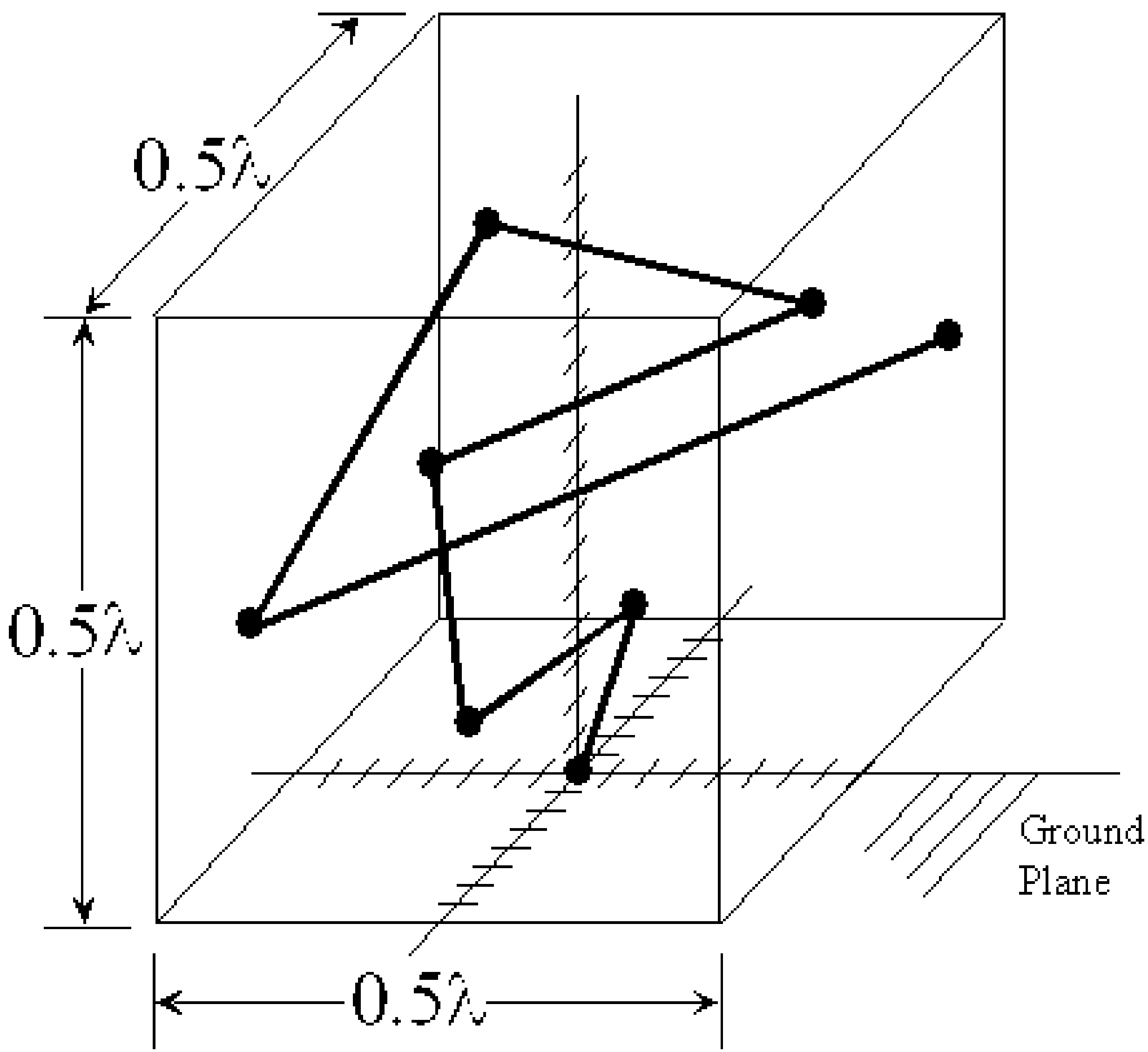
Num sections after smallest



- Middle section constrained to be smallest,
- That's all
- Mutations can change diameters, add sections, and delete sections

Some slides from an Introductory lecture by John Koza, Stanford

Antenna Design



Antenna Design

- The problem (Altshuler and Linden 1998) is to determine the *x-y-z* coordinates of the 3-dimensional position of the ends (*X1, Y1, Z1, X2, Y2, Z2,… , X7, Y7, Z7*) of 7 straight wires so that the resulting 7-wire antenna satisfies certain performance requirements

- The first wire starts at feed point (0, 0, 0) in the middle of the ground plane

- The antenna must fit inside the 0.5λ cube

Antenna Genome

X_1	Y_1	Z_1	X_2	Y_2	Z_2	…
+0010	**-1110**	**+0001**	**+0011**	**-1011**	**+0011**	**…**

- **105-bit chromosome (genome)**
- **Each *x-y-z* coordinate is represented by 5 bits (4-bit granularity for data plus a sign bit)**
- **Total chromosome is $3 \times 7 \times 5 = 105$ bits**

This was done with binary encoding – the fashion at the time, and still often used, but real-valued or integer coding will be just as applicable

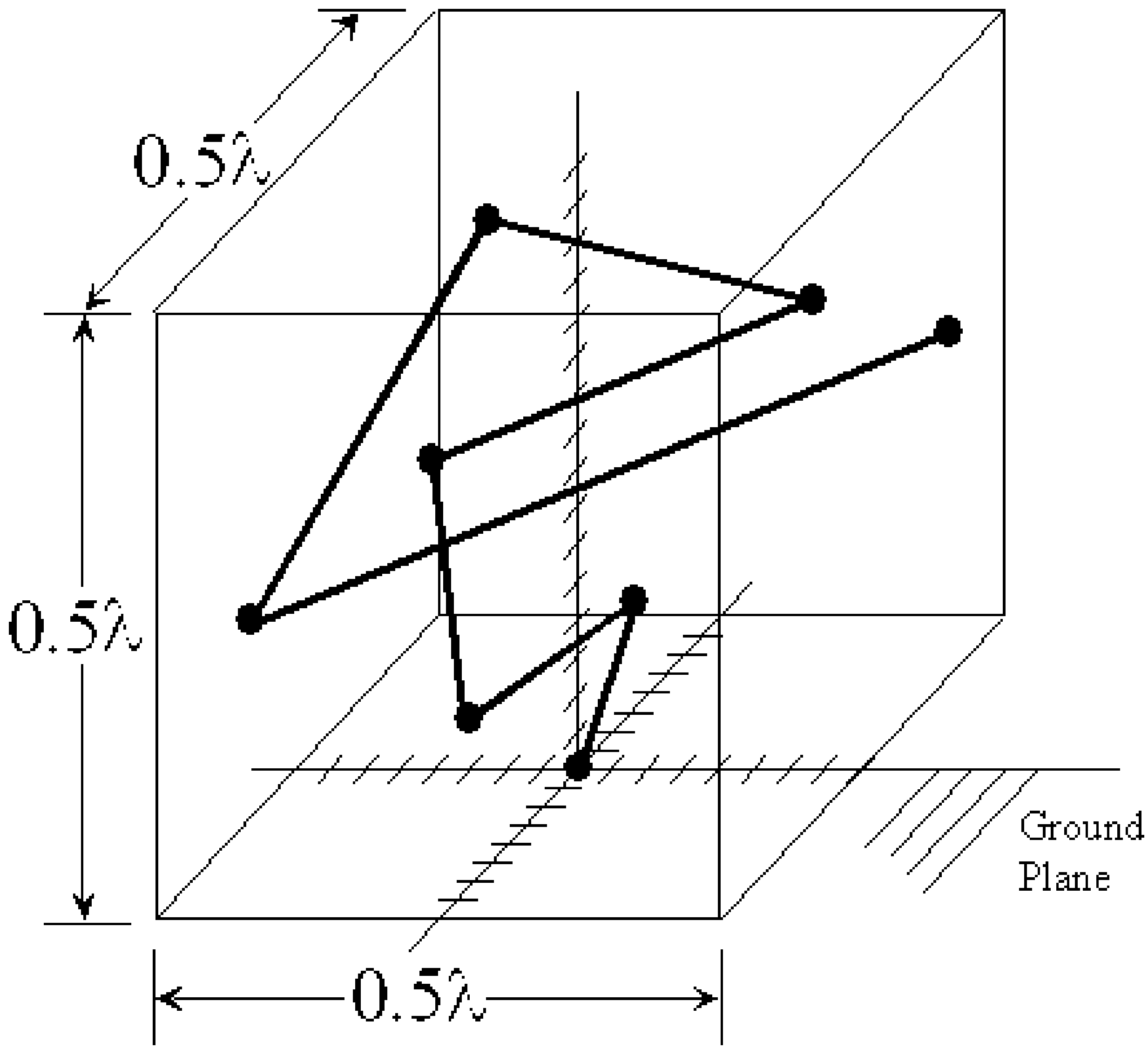
Antenna Fitness

- Antenna is for ground-to-satellite communications for cars and handsets
- We desire near-uniform gain pattern 10° above the horizon
- Fitness is measured based on the antenna's radiation pattern. The radiation pattern is simulated by National Electromagnetics Code (NEC)

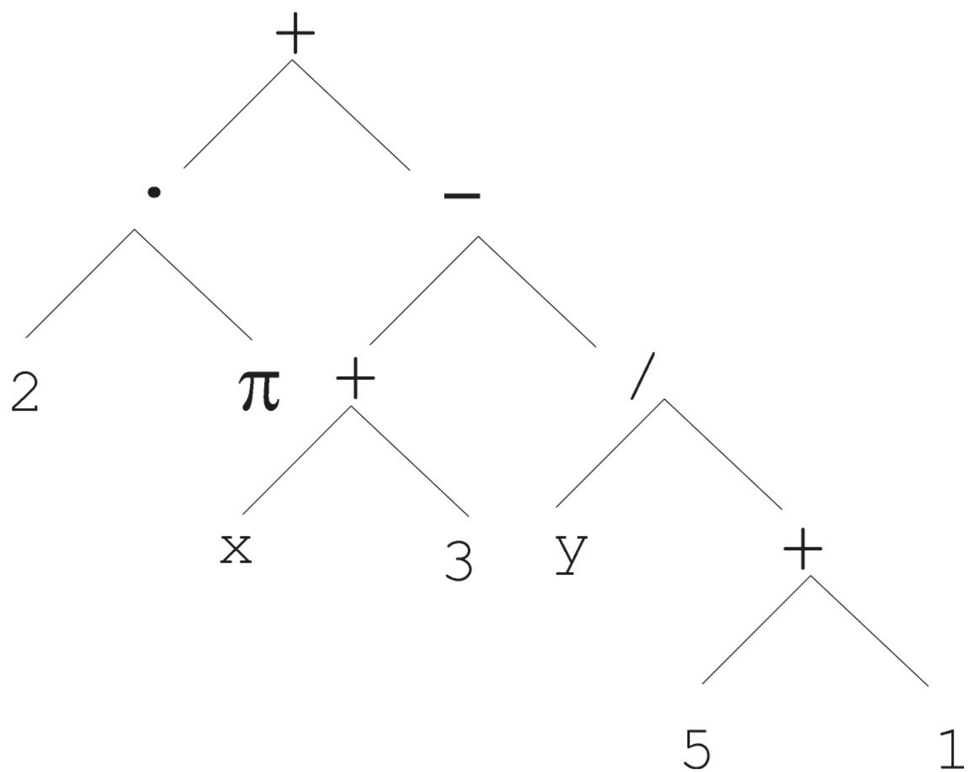
Antenna Fitness

- Fitness is sum of the squares of the difference between the average gain and the antenna's gain
- Sum is taken for angles Θ between -90° and +90° and all azimuth angles Φ from 0° to 180°
- The smaller the value of fitness, the better

U. S. PATENT 5,719,794



Tree based representation



$$2 \cdot \pi + \left((x + 3) - \frac{y}{5 + 1} \right)$$

Conclusions

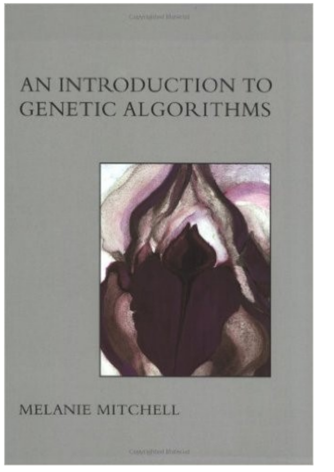
- The way a problem is encoded determines the:
  - The difficulty of the problem (fitness landscape)
  - Mutation operator
  - Crossover operator

- Encoding must be done taking into considerations the constraints of the problem

- Different types of encoding include:
  - Integer vectors
  - Real vectors
  - Bit & symbol vectors
  - Trees

- Each of these can be direct or indirect

- Readings:
  – An Introduction to Genetic Algorithms by M. Mitchell
    Chapter 2 (Problem Solving with EAs)



- Next Lecture: Genetic Programming